

UNIVERSITY PARTNER



6CS012 – Artificial Intelligence and Machine Learning

Assignment 1 – Question and Answer Report

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Table of Contents

1. Long Question	1
1.1. Unsupervised Learning Opportunities for eSewa	1
A. Customer Behaviour Segmentation	1
B. Fraud and Abnormal Transaction Detection	1
2. Short Questions	2
2.1. Overfitting and Underfitting	2
Overfitting.....	2
Underfitting.....	2
2.2. Neural Network Architecture: CNN and RNN	3
Common Training Challenges and Solutions	4
3. References	5

1. Long Question

Scenario: You are employed as a Data Scientist at eSewa, a major digital payment service in Nepal. Discuss two important business areas where unsupervised learning could add practical value. For each area, state the business issue, suggest a suitable unsupervised learning technique and algorithm, and describe how the results could be used within eSewa's products or operational services.

1.1. Practical Uses of Unsupervised Learning at eSewa

Unsupervised learning is especially useful for a payment platform like eSewa because it can identify hidden structures in large customer and transaction datasets without depending on manually labelled examples. This is important in situations where customer habits change regularly, fraudulent behaviour develops quickly, and business teams require clearer evidence about how users interact with the platform. Two valuable applications for eSewa are customer behaviour grouping and the detection of unusual or potentially fraudulent transactions.

A. Customer Behaviour Grouping and Segmentation

Problem definition. eSewa users show different patterns of platform usage. Some customers may make regular wallet payments, others may mainly use the app for utility bills, and some may only log in occasionally. Because these users have different needs, activity levels and spending behaviours, one general marketing approach is unlikely to work well for every customer group.

Recommended approach. A clustering technique can divide customers into groups based on similarities in their transaction activity. Possible input variables include how often users transact, their average payment amount, monthly spending level, preferred service categories, bill-payment behaviour, merchant-payment usage and period of inactivity. Principal Component Analysis (PCA) can be applied first to reduce dimensionality and make the data easier to interpret. K-Means clustering can then be used to form clear user segments. If the data contains irregular group shapes or many outliers, DBSCAN may be a stronger alternative.

Integration into eSewa services. The segments generated by the model can guide more personalised marketing and service design. For instance, highly active users could be offered loyalty benefits, inactive customers could receive re-engagement promotions, and users who mostly pay bills could be shown reminders or package-based offers. These customer insights can help eSewa improve campaign effectiveness, increase retention and prioritise features according to real usage behaviour.

B. Fraud and Unusual Transaction Detection

Problem definition. Fraud in digital payment platforms is challenging because suspicious behaviour can change over time. New fraud strategies may not be represented in older labelled datasets, which reduces the reliability of purely supervised detection systems. Suspicious activity is often visible as behaviour that differs strongly from a customer's normal pattern, such as a very large transaction, repeated transfers within a short period, or access from an unfamiliar device or location.

Recommended approach. Anomaly detection can be used to highlight transactions that differ significantly from usual behaviour. Autoencoders are appropriate because they can learn normal transaction patterns and then calculate reconstruction error for new transactions. A transaction with a

high reconstruction error can be treated as unusual and forwarded for investigation. Isolation Forest and One-Class SVM are also useful options for tabular transaction records and can support the same risk-detection process.

Integration into eSewa services. The anomaly scores produced by the model can be connected to eSewa's risk monitoring workflow. Transactions with high risk scores could be marked in real time, reviewed manually, or checked using extra security steps such as OTP confirmation or two-factor authentication. This would help reduce financial losses, maintain customer confidence and improve platform security while avoiding unnecessary blocks on normal customer activity.

Overall, these applications show how unsupervised learning can contribute to both business growth and platform protection. Segmentation supports better engagement and personalisation, while anomaly detection helps eSewa respond to changing fraud risks more effectively.

2. Short Questions

2.1. Overfitting and Underfitting

Question: Define overfitting and underfitting in machine learning and explain the differences between them. Describe why both conditions reduce model performance and include simple examples to support the explanation.

Overfitting and underfitting are two common modelling problems that affect how well a machine learning model works on unseen data. A strong model should capture the main relationship in the dataset without becoming too complex or too simple.

Overfitting

Overfitting happens when a model learns the training dataset too closely. It captures useful patterns but also memorises noise and small random details. Therefore, it can score highly on training data but perform poorly on new data.

Why it is problematic. An overfitted model has low bias and high variance. It depends too much on the training examples, so its predictions may become unreliable in real-world use.

Example. A very high-degree polynomial may pass through almost every training point. However, the curve can become overly irregular and may give inaccurate predictions for new points.

Underfitting

Underfitting occurs when a model is too simple to represent the main patterns in the dataset. It usually performs poorly on both training and testing data because it has not learned the real relationship well.

Why it is problematic. An underfitted model has high bias and low variance. It makes assumptions that are too simple and misses important structure, making it unsuitable for practical prediction tasks.

Example. A straight-line regression model will not fit data that follows a curved or parabolic pattern, so the model will produce large errors.

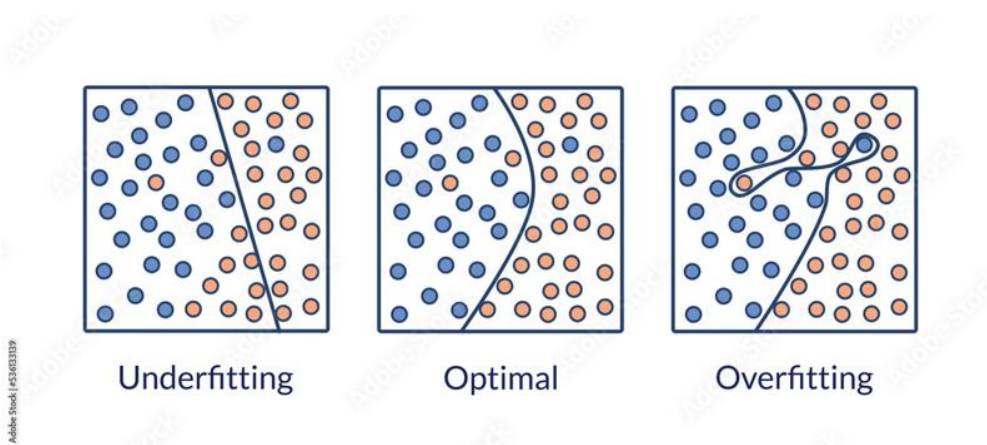


Figure 1: Visual comparison of underfitting, optimal fit and overfitting

In conclusion, underfitting means the model learns too little, whereas overfitting means it learns the training data in too much detail. The ideal model balances complexity and generalisation.

2.2. Neural Network Architecture: CNN and RNN

Question: Explain the key differences between Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs). Provide suitable examples of where each architecture can be used and briefly discuss common training challenges in deep learning with possible solutions.

Convolutional Neural Networks and Recurrent Neural Networks are both widely used deep learning models, but they are designed for different forms of data and different prediction tasks.

Aspect	Convolutional Neural Network (CNN)	Recurrent Neural Network (RNN)
Main purpose	Learns spatial features from grid-style data, especially images.	Learns patterns where the order of data points or time steps is important.
How it works	Applies convolution filters and pooling operations to identify local features such as edges, textures, shapes and objects.	Uses hidden states and recurrent connections so information from earlier steps can influence later outputs.
Best suited for	Image classification, object detection, face recognition and medical image interpretation.	Time-series forecasting, speech recognition, sentiment analysis and other natural language processing tasks.
Example	Classifying whether a picture shows a vehicle, a building or a person.	Forecasting sales, interpreting text sequences or identifying spoken words.

A CNN is most appropriate when the dataset contains spatial relationships. In image recognition, for example, neighbouring pixels are connected, and convolutional filters can identify useful patterns even if those patterns appear in different parts of the image. An RNN is more suitable when the sequence or order of information is important. In sentence analysis or time-series forecasting, the current input may depend on earlier words or previous values.

Common Training Challenges and Possible Solutions

Challenge	Explanation	Possible Solution
Vanishing gradients	During backpropagation, gradients may become very small, particularly in deep networks or basic RNNs, causing earlier layers to learn slowly.	Use LSTM or GRU models for sequence tasks, apply suitable weight initialisation and use batch normalization where appropriate.
Exploding gradients	Gradients may become excessively large, making training unstable and causing sudden increases in loss.	Use gradient clipping and select a suitable learning rate.
Overfitting	The model remembers the training data too closely and loses accuracy on unseen data.	Use dropout, early stopping, regularisation, data augmentation and a separate validation set.
Slow training	Large deep learning models can require substantial data, time and computational power.	Apply transfer learning, mini-batch training, GPU support and efficient model architectures.

Therefore, CNNs are usually selected for visual or spatial problems, while RNNs are better suited to sequential data. During training, problems such as vanishing gradients, exploding gradients, slow learning and overfitting should be managed through techniques such as LSTM/GRU units, gradient clipping, batch normalization, dropout, early stopping and transfer learning.

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